

# Week 9: Backtesting, Risk Analysis & Taker Strategies

Assembling the full system – Milestone 3 (Coincall guest session)

North Carolina State University | Summer 2026

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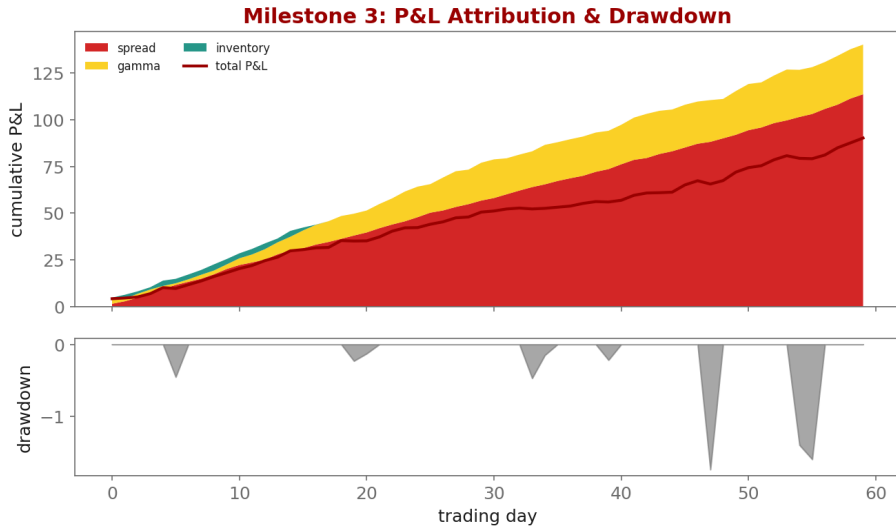
Industry Mentor: Dendi Suhubdy (Bitwyre) | Guest: Fenni Kang (Coincall)

1. Learning objectives
2. Lecture 1: Backtest Architecture & Risk
3. Lecture 2 (Guest: Fenni Kang, Coincall): Taker Strategies
4. Illustration
5. Derivation & rationale
6. Getting the data from Coincall
7. Code: week9\_attribution.py
8. Summary & key takeaways
9. Glossary of key terms
10. Reading
11. References

- ▶ Build an event-driven options backtest
- ▶ Produce a full P&L attribution and risk metrics
- ▶ Understand taker strategies and order-flow toxicity

- ▶ Event-driven simulation with deterministic replay
- ▶ Fill simulator; state of inventory and Greeks
- ▶ Pitfalls: look-ahead, unrealistic fills, expiry handling
- ▶ Sharpe with block-bootstrap confidence intervals

- ▶ When and why takers cross the spread
- ▶ Signals behind aggressive flow: momentum, liquidations, news
- ▶ What taker behaviour means for the maker's toxicity



$$\widehat{SR} = \frac{\bar{r}}{s_r} \sqrt{252}, \quad \text{CI via block bootstrap on } \{r_t\}$$

**Why:** Daily Sharpe is annualized by  $\sqrt{252}$ , but a point estimate over a few months is noisy and serially correlated. We resample BLOCKS of returns (preserving autocorrelation) to get an honest confidence interval – a high Sharpe with a wide CI is not a real edge.

- ▶ Full options L2 + trades + perp + funding for 3 months:
  - ▶ `coincall_btc_options_l2_2025Q4.parquet / ..._trades...`
  - ▶ `coincall_eth_options_l2_2025Q4.parquet / ..._trades...`
  - ▶ `coincall_btc_eth_perp_l2_2025Q4.parquet ; coincall_funding_rates...`
- ▶ Klines for regime context: GET  
`/open/option/market/kline/history/v1/{name}?period=h1`
- ▶ Trade tape carries mark IV at trade -> calibrate the fill simulator

*Cross-check exact paths at <https://docs.coincall.com/>*

```

import numpy as np
def attribute(spread, inventory, gamma, hedging):
    total = spread + inventory + gamma + hedging
    assert abs(total.sum() - (spread.sum()+inventory.sum()+
                              gamma.sum()+hedging.sum())) < 1e-6 # must sum to total
    return dict(spread=spread, inventory=inventory, gamma=gamma,
                hedging=hedging, total=total)

def bootstrap_sharpe(r, block=5, n=2000, seed=0):
    rng = np.random.default_rng(seed); T = len(r); out = []
    for _ in range(n):
        idx = (rng.integers(0, T-block, T//block)[: ,None] + np.arange(block)).ravel()
        s = r[idx]; out.append(s.mean()/s.std()*np.sqrt(252))
    return np.percentile(out, [2.5, 50, 97.5]) # CI for the Sharpe

```

- ▶ Event-driven, deterministic replay; beware look-ahead and 100% fills
- ▶ Attribution MUST sum to total P&L – a mismatch is always a bug
- ▶ Report Sharpe with a block-bootstrap CI, not a bare point estimate
- ▶ Taker flow (guest lecture) shapes the toxicity the maker must price
- ▶ Milestone 3: full backtest on 3 months BTC & ETH, with a drawdown forensic

**Event-driven backtest** a simulation that replays timestamped events in order

**Deterministic replay** bitwise-identical results on re-run

**Look-ahead bias** using information not available at decision time

**Sharpe ratio** risk-adjusted return: annualized mean over standard deviation

**Block bootstrap** resampling blocks of returns for a CI that keeps autocorrelation

**Maximum drawdown** the largest peak-to-trough decline in equity

**Taker strategy** aggressive, spread-crossing execution

- ▶ Cartea et al., Ch. 10-11; Bailey-Lopez de Prado (2014); Lopez de Prado (2018)

- ▶ Cartea, Jaikumar, & Penalva (2015). Algorithmic and High-Frequency Trading, Ch. 10-11.
- ▶ Bailey, D. H., & Lopez de Prado, M. (2014). The deflated Sharpe ratio. *Journal of Portfolio Management*, 40(5), 94-107.
- ▶ Lopez de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
- ▶ Coincall API documentation. <https://docs.coincall.com/>